

Samuel Karanja

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Current role	Lead Developer, Violence Detection System at Eyeknow and Senior Business Process Automation Specialist at Freelance
Technologies	Python, Tensorflow, Keras, Salesforce, Hubspot, Zapier, PyTorch, SQL

Work experience

Lead Developer, Violence Detection System, Eyeknow 🏢

Jan 2022 - Present (4y 2m)

Python Tensorflow Keras

- Developed and implemented a real-time violence detection system with deep learning models, achieving a 90% accuracy rate in live monitoring.
- Designed and automated an event detection system that seamlessly integrates with security procedures, enabling quick alerts to security teams and decreasing incident response times.
- Improved model inference and streaming efficiency for low-latency processing across multiple CCTV feeds using Python, OpenCV, and GPU acceleration.

Senior Business Process Automation Specialist, Freelance 🏢

Jan 2019 - Present (7y 2m)

Salesforce Hubspot Zapier

- Designed and implemented over 50 automation solutions using Zapier, Make.com, n8n, and Google Apps Script, resulting in clients saving an average of 25 hours per week.
- Developed advanced integrations for CRM and marketing automation via REST APIs, webhooks, and platforms such as HubSpot and Salesforce, improving lead management and conversion tracking effectively.
- Implemented data-focused automation to enhance reporting, notifications, and approvals, increasing operational capacity by over 20% for clients with streamlined processes and reduced manual tasks.

Lead Machine Learning Engineer, Weedbot 🏢

Jan 2022 - Jan 2023 (1y)

Python Tensorflow PyTorch

- Led a 40-person global team in creating a computer vision-based laser weeding system for agricultural automation.
- Developed a real-time image recognition pipeline using CNN and LSTM for crop-weed classification achieving 98% model accuracy in field conditions.
- Automated Python data workflows for data ingestion, feature engineering, and batch processing, decreasing manual intervention by 80%.

SQL

Python

Tensorflow

- Developed real-time object recognition model to identify buses internationally, assisting visually impaired users.
- Implemented OCR pipeline for precise extraction of bus numbers and routes, enabling independent bus verification by users.
- Designed reliable object detection system to locate empty seats, improving accessibility and safety for visually impaired passengers.

Education

World Quant University

Oct 2025 - Jan 2026 (3m)

MSc in Financial Engineering

Kenya Methodist University

May 2018 - Nov 2022 (4y 6m)

BSc Mathematics & Computer Science